3. 3절

1(2)

$$2y'' - 5y' = 0$$

$$2m^2 - 5m = 0$$

$$2m\left(m - \frac{5}{2}\right) = 0$$

$$m = 0, \ \frac{5}{2}$$

$$\therefore y = c_1 e^{0x} + c_2 e^{\frac{5}{2}x}$$

$$= c_1 + c_2 e^{\frac{5}{2}x} \quad //$$

3. 4절

1(1)

$$y'' + 3y' + 2y = 6$$

$$m^2 + 3m + 2 = 0$$

$$m = -1, -2$$

$$\therefore y_c = c_1 e^{-x} + c_2 e^{-2x}$$

$$y_p = a$$

$$2a = 6$$

$$\therefore a = 3$$

$$\therefore y_c + y_p$$

$$= c_1 e^{-x} + c_2 e^{-2x} + 3$$

2(3)

$$y'' + 6y' + 5y = 0, \ y(0) = 0, \ y'(0) = 3$$

$$m^2 + 6m + 5 = 0$$

$$m = -1, -5$$

$$\therefore y_c = c_1 e^{-x} + c_2 e^{-5x}$$

$$y(0) = c_1 + c_2 = 0$$

$$y'(0) = -c_1 - 5c_2 = 3 \quad //$$

$$-4c_2 = 3, \ c_2 = -\frac{3}{4}, \ c_1 = \frac{3}{4}$$

$$\therefore y = \frac{3}{4} e^{-x} - \frac{3}{4} e^{-5x}$$

$$//$$

2(2)

$$2y'' + 3y' - 2y = 14x^2 - 4x - 11,$$

$$y(0) = 0, \ y'(0) = 0$$

$$2m^2 + 3m - 2 = 0$$

$$(2m - 1)(m + 2) = 0$$

$$m = \frac{1}{2}, \ -2$$

$$y_c = c_1 e^{\frac{1}{2}x} + c_2 e^{-2x}$$

$$y_p = ax^2 + bx + c$$

$$y_p' = 2ax + b$$

$$y_p'' = 2a$$

$$W(e^{\frac{1}{2}x}, e^{-\frac{1}{2}x}) = \begin{vmatrix} e^{\frac{x}{2}} & e^{-\frac{x}{2}} \\ \frac{1}{2}e^{\frac{x}{2}} & -\frac{1}{2}e^{-\frac{x}{2}} \end{vmatrix}$$

$$= -\frac{1}{2} - \frac{1}{2} = -1$$

$$u_1' = + \underbrace{e^{-\frac{1}{2}x} \cdot \frac{1}{4}x \cdot e^{\frac{x}{2}}}_{+1}$$

$$= \frac{1}{4}x$$

$$u_2' = \underbrace{e^{\frac{x}{2}} \cdot \frac{1}{4}x \cdot e^{\frac{x}{2}}}_{1}$$

$$= \frac{1}{4}x \cdot e^{x}$$

$$u_1 = \frac{1}{8}x^2$$

$$u_2 = \frac{1}{4}x \cdot e^{x} - \frac{1}{4}e^{x}$$

$$\therefore y_p = \frac{1}{8}x^2 e^{\frac{1}{2}x} + \underbrace{\frac{1}{4}e^{x}(x-1) \cdot e^{-\frac{1}{2}x}}_{\frac{1}{4}(x-1)e^{\frac{x}{2}}}$$

$$= e^{\frac{x}{2}}\left(\frac{1}{8}x^2 + \frac{1}{4}x - \frac{1}{4}\right)$$

$$y_c + y_p = c_1 e^{\frac{1}{2}x} + c_2 e^{-\frac{1}{2}x} + e^{\frac{x}{2}}\left(\frac{1}{8}x^2 + \frac{1}{4}x - \frac{1}{4}\right)$$

$$y(0) = c_1 + c_2 - \frac{1}{4} = 1$$

$$y'(0) = \frac{1}{2}c_1 - \frac{1}{2}c_2 + \frac{1}{2}\cdot\frac{1}{4} = 0$$

$$\begin{cases} c_1 + c_2 = \frac{5}{4} \\ c_1 - c_2 = -\frac{1}{4} \end{cases} \qquad 2c_1 = 1$$

$$c_1 = \frac{1}{2}, \quad c_2 = \frac{3}{4}$$

$$\therefore y = \frac{1}{2}e^{\frac{1}{2}x} + \frac{3}{4}e^{-\frac{1}{2}x} + e^{\frac{x}{2}}\left(\frac{1}{8}x^2 + \frac{1}{4}x - \frac{1}{4}\right)$$

3.6절

#1(2)

$$4x^2y'' + y = 0$$

$$x^m\left(4m(m-1) + 1\right) = 0$$

$$\underbrace{4m^2 - 4m + 1 = 0}_{(2m-1)^2}$$

$$m = \frac{1}{2}\ (\text{중근})$$

$$\therefore y_c = c_1 x^{\frac{1}{2}} + c_2 x^{\frac{1}{2}} \cdot \ln x$$

#3(1)

$$xy'' + y' = x$$

$$x \cdot \left(m(m-1) \cdot x^{m-2}\right) + m x^{m-1} = 0$$

$$\therefore m^2 = 0$$

$$\therefore y_c = c_1 x^0 + c_2 x^0 \ln x$$

$$= c_1 + c_2 \ln x$$

$$y'' + \frac{y'}{x} = 1$$

$$\therefore \quad 4a + 6ax + 3b - 2ax^2 - 2bx - 2c = 14x^2 - 4x - 11$$

$-2a = 14 \quad \cdots \quad a = -7$

$6a - 2b = -4 \quad \therefore \quad b = -19$

$4a + 3b - 2c = -11 \quad c = -37$

$-28 - 57$

$$\therefore \quad y_p = -7x^2 - 19x - 37$$

$$y_c + y_p = c_1 e^{\frac{1}{2}x} + c_2 e^{-2x} - 7x^2 - 19x - 37$$

$$y(0) = c_1 + c_2 - 37 = 0$$

$$y'(0) = \frac{1}{2}c_1 - 2c_2 - 19 = 0$$

$$\begin{cases} c_1 + c_2 = 37 \\ c_1 - 4c_2 = 38 \end{cases}$$

$$c_2 = -\frac{1}{5}, \quad c_1 = \frac{186}{5}$$

$$\therefore \quad y = \frac{186}{5} e^{\frac{1}{2}x} - \frac{1}{5} e^{-2x} - 7x^2 - 19x - 37$$

3.5절

#1(1)

$$y'' + y = \sec x$$

$$m^2 + 1 = 0$$

$$m = \pm i$$

$$y_c = c_1 \cos x + c_2 \sin x$$

$$W(\cos x, \sin x) = \begin{vmatrix} c & s \\ -s & c \end{vmatrix} = 1$$

$$\therefore \quad u_1' = -\frac{\sin x \cdot \sec x}{1}$$

$$= -\tan x$$

$$u_2' = \frac{\sin x \cdot \cos x}{1}$$

$$= \sin x \cos x$$

$$u_1 = \int -\tan x \, dx$$

$$= \int \frac{1}{t} \, dt$$

$$= \ln |\cos x|$$

$$u_2 = \frac{1}{2}\sin^2 x$$

$$\therefore \quad y_p = \ln |\cos x| \cdot (\cos x) + \frac{1}{2}\sin^3 x$$

$$\therefore \quad y = c_1 \cos x + c_2 \sin x + \cos x \cdot \ln |\cos x| + \frac{1}{2}\sin^3 x,$$

$$(-\infty < x < \infty)$$

#2(1)

$$y(0) = 1, \quad y'(0) = 0$$

$$4y'' - y = x \cdot e^{\frac{x}{2}}$$

$$y'' - \frac{1}{4}y = \frac{1}{4}x \cdot e^{\frac{x}{2}}$$

$$m^2 = \frac{1}{4}$$

$$m = \pm \frac{1}{2}$$

$$y_c = c_1 e^{\frac{1}{2}x} + c_2 e^{-\frac{1}{2}x}$$

3장 연습문제

3.1절

4

$y = c_1 + c_2 \cos x + c_3 \sin x$

$y(\pi) = 0$

$y'(\pi) = 2$

$y''(\pi) = -1$

$c_1 - c_2 = 0 \quad \cdots ①$

$y' = -c_2 \sin x + c_3 \cos x$

$y'(\pi) = -c_3 = 2 \quad \cdots ②$

$y''(\pi) = c_2 = -1 \quad \cdots ③$

$\therefore c_1 = -1, \ c_2 = -1, \ c_3 = -2$

$\therefore y = -1 - \cos x - 2\sin x$ //

5

$y = c_1 x + c_2 x \ln x$

$y' = c_1 + c_2(\ln x + 1)$

$y(1) = 3, \quad y'(1) = -1$

$y(1) = c_1 = 3 \quad \cdots ①$

$y'(1) = c_1 + c_2 = -1 \quad \cdots ②$

$\therefore c_1 = 3, \ c_2 = -4$

$\therefore y = 3x - 4x \ln x$ //

3.2절

5(2)

$y'' - 4y = 0,$

$\cosh 2x,$

$\sinh 2x,$

$(-\infty, \infty)$

$m^2 - 4 = 0$

$m = \pm 2$ //

$\therefore$ (일반해)

$= y = c_1 e^{2x} + c_2 e^{-2x}$ //

$\therefore$ 해의 기본계를 이룬다.

9(4)

$y'' + 2y' + y = 0, \quad y_1 = x \cdot e^{-x}$

$y_2 = y_1 \int \frac{e^{-\int p\,dx}}{y_1^2}\,dx$

$= x \cdot e^{-x} \int \frac{e^{-2x}}{x^2 e^{-2x}}\,dx$

$= x \cdot e^{-x} \cdot \left(-\frac{1}{x}\right)$

$= -e^{-x}$ //

$$y_1 = 1, \quad y_2 = \ln x$$

$$W = \begin{vmatrix} 1 & \ln x \\ 0 & \frac{1}{x} \end{vmatrix}$$

$$= \frac{1}{x}$$

$$u_1' = -\frac{\ln x \cdot 1}{\frac{1}{x}} = -x \ln x$$

$$u_2' = \frac{1 \cdot 1}{\frac{1}{x}} = x$$

$$\int +x \ln x \, dx$$

$$= \ln x \cdot \tfrac{1}{2}x^2 - \int \tfrac{1}{2}x^2 \cdot \tfrac{1}{x} dx \quad \nearrow \frac{x}{2}$$

$$= \ln x \ \tfrac{1}{2}x^2 - \frac{x^2}{4}$$

$$u_1 = \frac{x^2}{4} - \frac{1}{2}x^2 \ln x$$

$$u_2 = \frac{1}{2}x^2$$

$$y_p = \frac{x^2}{4} - \cancel{\frac{1}{2}x^2 \ln x} + \cancel{\frac{1}{2}x^2 \ln x}$$

$$y = c_1 + c_2 \ln x + \frac{x^2}{4} \quad //$$